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The effects of location-based tax policies on the distribution of household income: Evidence from the federal Empowerment Zone program

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ABSTRACT

Location-based tax policies are redistributive as evidenced by their placement in distressed areas. However, the previous literature has focused on mean effects which can mask important effects that the program has on the distribution of households. Therefore, we extend the literature by studying changes in the entire household income distribution, in the context of the federal Empowerment Zone (EZ) program. We do not find evidence that the impoverished residents benefited from the program. Our findings are consistent with the areas becoming more attractive to high-income households. The improvements in the areas were concentrated in those portions of each zone that were relatively better-off prior to EZ designation. The results confirm the prior literature findings that the areas, on average, became more attractive but also suggest that the benefits of the program likely did not accrue to the lower-income residents of the EZ areas.

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1. Introduction

Policymakers are increasingly relying on location-based tax policies to help their constituents spurring a proliferation of studies that attempt to determine their effectiveness. However, despite the potential redistributive nature of these policies researchers have predominately studied the policy effects at the mean instead of throughout the income distribution. To investigate the importance of studying distributional changes in the context of location-based tax policies we study the federal Empowerment Zone (EZ) program, which offers a set of tax incentives to firms if they locate their business in specified distressed areas and hire workers who reside in the area. Specifically, the EZ program offers wage and capital tax incentives as well as service block grants to the community in order to induce businesses to locate in specified distressed areas. The largest and most utilized incentive (Hanson, 2011), the wage tax credit, provides employers (or employer) tax credits worth 20% of the wages of employees who live in these distressed areas, up to $3000. Overall, this program is the largest location-based redevelopment program evident by the Department of Housing and Urban Development (HUD) estimated annual cost of $11 billion.[1]

While the program was designed to stimulate economic activity in distressed areas, which could manifest in a variety of possible outcomes or metrics, at least part of the impetus for the program was to improve the lives of people living in the areas. In a 2002 joint letter to President George W. Bush, Senator Rick Santorum and Congressman J.C. Watts, Jr. stated that the goal of the EZ program is

> "...to create an environment that enables distressed urban and rural communities to have hope for the future through economic and social renewal. Our belief is that when private industry flourishes in these communities, it directly, and positively, impacts peoples' lives."[2]

Researchers interested in how the federal EZ program, or other economic development programs, impacts individuals' lives typically study socioeconomic indicators such as the poverty rate

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[1] The HUD estimate does include costs pertaining to the less generous and smaller Renewal Community and Enterprise Community areas which are not included in this paper.

[2] See U.S. Department of Housing and Urban Development, "Questions and Answers on Renewal Community and Empowerment Zone Tax Incentives," at http://www.hud.gov/offices/cpd/economicdevelopment/programs/rc/index.cfm.

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(e.g. Oakley and Tsao, 2006, Hanson, 2009; Ham et al., 2011), income (e.g. Hanson, 2009; Krupka and Noonan, 2009; Busso et al., 2013), or housing values (e.g. Hanson, 2009; Krupka and Noonan, 2009; Busso et al., 2013). To evaluate whether or not the EZ program improves the lives of people residing in these impoverished areas, we propose to investigate changes in the income distribution that would suggest that households are being moved out of poverty.

Estimated effects on mean and median income or the overall poverty rate, as have been produced in the prior literature, may provide little information about this question. In fact, evidence about the success of the federal EZ program is mixed.[3] For example, Ham et al. (2011) find large increases in average income, Oakley and Tsao (2006) find no effect on average income, and Busso et al. (2013) find positive effects on weekly wages but not on average earnings per worker. Likewise, the EZ program's effect on the poverty rate has been found to be positive by Ham et al. (2011) while Krupka and Noonan (2009) found mixed results, and the U.S. General Accounting Office (2006), Hanson (2009) and Oakley and Tsao (2006) find no effect of the program. While some of the differences in estimates within the literature are due to methodological choices, a possible partial explanation is that the program has a complicated effect on the distributions of households across economic measures. These distributional effects could in turn affect mean impacts, potentially differently based on empirical methodology.

For illustration, suppose that the federal EZ program causes firms to hire some workers who are very low income and that doing so moves these households from severe to moderate poverty. This would have no effect on the estimated poverty rate, since all of the movement is below the poverty line, and could have little effect on the estimates of either median or mean household income, the latter because the distribution of incomes is highly skewed. Such estimates of the average effect could lead researchers and policymakers to conclude that the program has little overall effect on reducing poverty. However, investigation of the changes in the poverty distribution would suggest that the program was successful.

Alternatively, suppose instead that the EZ program causes an increase in the density of households at the top of the income distribution. This could result because the program attracted higher income individuals to reside in the area. This could produce a positive estimated effect of the program on mean income. If the researcher only observes the positive mean income effect, it could lead to incomplete or inaccurate policy conclusions. The result could be misconstrued as a positive because of increasing average incomes of residents of the EZ area, even though the target impoverished population has not benefited from the program. Of course, there are other possibilities for how the EZ program affects the distribution of economic outcomes but the larger point is that understanding the distributional impacts could be critical for policy analysis, and has largely not been considered previously in the literature on place-based policies.

We investigate how the federal EZ program affects the density of individuals and households across the income distribution, as well as the distributions of other previously considered outcomes such as housing costs. Our approach is similar to previous studies in other contexts, such as Neumark et al. (2005) who study how the distribution of family incomes is impacted by minimum wages and find no evidence of an increased density of households just above the poverty line. In our analysis, we use block group data from the 1980, 1990 and 2000 censuses. While the prior literature has used this data, or the more aggregated tract-level data, on average incomes and wages, we utilize information on the counts of households and individuals within specified ranges of household income. These counts allow us to construct estimates of the density of individuals and households across the income distribution, as well as other distributions, in each year of our data. Thus, we can estimate how the densities of individuals and households changes at various points in these distributions, thereby providing a more complete picture of the effects of the program compared to average estimates.[4]

For identification, we utilize the selection process of the program that chose distressed areas from a set of qualified applicants. Specifically, we compare changes in the distribution in areas that received EZ designation with areas that qualified and applied for the EZ program but were rejected. While this control group has been utilized frequently in the previous literature, there is evidence that the EZ program selected areas with worse observable characteristics (Ham et al., 2011; Busso et al., 2013; Reynolds and Rohlin, 2014). Similar to Busso et al. (2013), we account for these differences by reweighting the control group to be more similar to the areas selected for the EZ program. For each outcome we consider, we use weights based on the propensity score to construct a set of control areas that have the same empirical distribution of households or individuals, in both 1980 and 1990, as those areas ultimately selected for the EZ program in 1994. We will demonstrate that our procedure produces a control group with nearly identical empirical distributions in both 1980 and 1990, and therefore changes in the pre-designation distributions, compared to the EZ areas.

Overall, we find that EZ designation has a polarizing effect on the distribution of households relative to the poverty line. Estimates show an increase in the density of households making twice the poverty rate as well as an increase in the density of households whose income is less than half the poverty rate. We further find that the increase in households with income more than twice the poverty rate is driven by an increase in households making more than $100,000. This increase at the upper part of the income distribution explains at least part of the positive mean income effects of the EZ program found in the previous literature. We demonstrate that our results are robust to concerns discussed in the previous literature including the use of alternative matching methodologies, the use of imputed lagged values to avoid division bias (Borjas, 1980), and comparisons to areas bordering the Empowerment Zones.

These results provide no evidence that the existing impoverished residents experienced gains from the EZ program. First, there appears to be an increase in the severely impoverished in the EZ areas compared to the control group. Second, the increased density above $100,000 occurs at roughly four times the pre-designation average household income in EZ areas. While there could be wage and employment gains associated with the program, either through the $3000 wage credit or other program components described later, it seems unlikely that such gains could move the existing impoverished households so high up into the income distribution. Instead, the change in the distribution is more likely to be explained by the EZ areas becoming more attractive to higher income households. In this case, positive movements of mean income or poverty rates should not be interpreted as improvements to the targeted impoverished population.

[3] This could be due to a number of potential reasons. There have been numerous methodologies employ across a variety of outcomes including business- and employment-related outcomes which are outside the focus of this paper. Additionally, there is a large literature on state placed-polices which contribute to the discussion on place-based policies.

[4] This paper is not the first to consider the possibility that mean estimates may disguise underlying effects that are important for program evaluation. Bondonio and Greenbaum (2007) separate the mean effect of state EZ programs into gross flows of firms into and out of existence finding that mean estimates on firms counts miss important dynamics. Bitler et al. (2006) document important heterogeneous effects of welfare reform missed by more standard mean estimates.

Further evidence that the EZ areas have become more attractive to high-income households is found by looking at changes in other distributions within the EZ area. For example, we find increases in educational attainment, including increases in the proportion of the population with a college or graduate degree, that are unlikely to be caused by any wage or employment increases among the existing EZ residents due to the program. Additionally, we find changes in the distributions of housing costs, with an increase in high rental rates and an increase in housing prices that extends into the upper part of the distribution. Furthermore, despite the increase in housing costs, we find evidence that households are spending less of their income on housing following the program, consistent with higher-income individuals moving into relatively low-cost areas. While these results do suggest that the areas became more attractive, which is one metric by which the program could evaluated, there is little evidence that the benefits are accruing to the impoverished population previously living in these areas.

Finally, we present evidence of a spatial variation in the effect of the EZ program on poverty reduction within EZ areas, and this variation provides a further explanation for the polarizing effects we find on the poverty rate. In particular, we find that poverty improved in the portion of each EZ that was less impoverished in 1990 prior to the program. In contrast, the portion of each EZ area that was initially more impoverished shows no improvement in poverty, and point estimates actually suggest that poverty worsened. Thus, any benefit of the EZ program appears to be concentrated in those portions of the areas that were initially better off.

The following section describes the federal Empowerment Zone program and Section 3 presents our data and empirical methodology. Section 4 begins with a presentation of mean and median estimates of the program effects, similar to the prior literature, before presenting our estimates of the distributional effects. We provide a discussion and interpretation of these results in Section 5 and the last section concludes.

2. The federal EZ program

Faced with economically depressed areas, federal and state officials have instituted myriad location-based economic development policies. This proliferation has spawned many different types of incentive programs, such as investment tax credits (Chirinko and Wilson, 2008), tax increment financing (Dye and Merriman, 2000), and location-based tax incentives to name a few. These types of programs typically focus on attracting or retaining businesses in these areas by offering financial assistance, often in the form of tax relief. The hope of these policies is that by luring business activity to these blighted areas they offer economic opportunities to the local residents and bring back economic prosperity.

These policies tend to be redistributive in nature as the policies benefit some people directly or indirectly as well as potentially inducing individuals and businesses to relocate across geography. While the specific effects of such policies could vary from one program to the next, we focus our study on the federal Empowerment Zone program which was created as a part of the 1993 Omnibus Budget Reconciliation Act (OBRA, 1993, P.L. 103-66) to help improve the most distressed areas in the U.S. by offering a set of tax incentives to firms that operate inside specific zones. The Department of Housing and Urban Development chose from a pool of applicants that had met the requirements that 20% of the residents lived in poverty and have an unemployment rate of at least 6.3%.[5] The largest and most utilized tax incentive that the EZ program offers is a 20% employee wage tax credit up to $3000 to businesses located within the zone for each worker that lives in the zone. The tax package also offers some capital investment incentives by giving firms additional capital expensing for a larger set of purchases. Lastly, each awardee was granted $100 million in social service block grants to subsidize things such as counseling, day care for children, employment services, legal services and substance abuse recovery for residents in the zone. For full details of the EZ incentives see Hanson (2009).

We chose to study the federal EZ program for a number of reasons. First, the EZ program is one of the largest location-based economic development programs in the United States. Although we focus on the initial wave of recipients (Atlanta, Baltimore, Camden/Philadelphia, Chicago, Detroit, and New York City), the program has since expanded. Following the first round, two urban ECs received EZ status, 21 new urban EZs were designated and 7 new rural EZs were created (GAO, 2004). Second, because this is a national program we can study how a uniform set of incentives can impact different areas across the country, although there is some evidence that cities utilized the Social Service Block Grants in different ways (Oakley and Tsao, 2006; U.S. General Accounting Office, 2006).[6] Additionally, the federal EZ program has been a blueprint of some of the over 40 state location-based economic development programs (Kenyon, 2012).[7]

Third, the application process provides comparison areas that allow us to build counterfactuals by using the fact that the recipients were chosen from a group of qualified applicants. Similar to some of the previous EZ literature (e.g. Hanson, 2009; Hanson and Rohlin, 2011a, 2011b; Busso et al., 2013), we use areas that applied and qualified for the program but were not ultimately chosen as a basis for constructing counterfactuals for the areas that did receive the full benefits of the program. In particular, we utilize a propensity score matching approach, described below, to construct a counterfactual from these areas that were not selected that have similar pre-1994 trends in outcomes as the EZ areas that were actually selected, similar to Busso et al. (2013). Most of the areas that were not chosen did receive "Enterprise Community" (EC) designation which provided them some of the capital incentives and $3 million in Social Service Bock Grants but were not allowed to claim the generous wage tax credit. A final useful feature of the program is that the boundaries of both the recipients and the rejected applicants are clearly defined according to census tracts and recorded by HUD.

3. Data and empirical methodology

Suppose that we are interested in estimating the economic effect of EZ designation as measured by some outcome θ. While the prior literature has generally focused on estimating the effect on some specific moment of θ, such as the average $\bar{\theta}$, we are

[5] There is a rural component to the program which chose three rural areas (Kentucky Highlands, Mississippi Delta, and the Rio Grande Valley in Texas). These areas are not included in our analysis because they are substantially different to the urban areas and an entirely different department, the Department of Agriculture, chose the areas that received the funding. This restriction to urban areas is common to the literature that has studied the federal EZ program (e.g. Hanson, 2009; Hanson and Rohlin, 2013; Busso et al., 2013).

[6] For instance, Baltimore's plan consisted of eight different components while Chicago's plan was comprised of only two components. See Oakley and Tsao (2006) for a more detailed explanation of the differences between each city. Although not directly studied in this paper, our results are supportive of the idea that the EZ program has differential effects across cities, which is a potential area for future research.

[7] There is a large literature that studies the effect of these state economic development programs. However, there does not appear to be a consensus on their impact. Papke (1994), O'Keefe (2004), Couch et al. (2005), Billings (2008), and Freedman (2013) find positive effects from state programs while others such as Boarnet and Bogart (1996), Bondonio and Engberg (2000), Faulk (2002), Greenbaum and Engberg (2004), Bondonio and Greenbaum (2007), Elvery (2009) and Neumark and Kolko (2010) find small or no net effects.

interested in how EZ designation changes the distribution of θ across individuals and households within EZ areas. The main concern is that estimating average effects may miss important changes in the underlying distribution and those changes in the distribution can be informative for drawing policy conclusions. This may be particularly true in the case of location-based tax policies as the effects of programs could: (1) vary across agents within the area, (2) could vary geographically within the area depending on the distribution of individuals and firms across space, and (3) could cause individuals and firms to move into or out of the targeted location, altering the observed distribution within the area.

To investigate these issues we use block group data from the Summary 3 Files of the 1980, 1990 and 2000 censuses, which are both publicly available and have been used frequently in the prior EZ literature. These files provide census data on individuals and households aggregated to various geographic levels, including block groups and tracts, and we use information from the 838 EZ block groups and 3659 EC block groups.[8] However, while the prior literature has focused on outcomes such as the poverty rate, wage and salary income and the unemployment rate, we will focus on related outcomes for which information has been provided about the distribution of the variable within the block group. For example, for each block group the data reports population counts within specified ranges of the ratio of household income to the poverty line. In particular, the data reports the number people living in the block group whose household income is less than 50% of the poverty line, more than 200% of the poverty line, and various ranges between. While we are limited to the ranges provided in the data, these counts can be used to construct an estimate of the density of households across the distribution of income relative to needs within the block group.

In our main analysis we focus on the density of individuals and households across four outcomes (θ) for which we have information about distributions: household income relative to the poverty line, household income, gross rent and house values. While the poverty rate, gross rent and house values have been previously considered in the literature, the prior studies have tended to focus more on wage and salary income as opposed to household income because that measure is tied more closely to the wage credit component of the EZ program. Unfortunately, information on the distribution of wage income is only provided in 2000 so we are unable to analyze changes in the distribution of wage income. Instead we use household income which is also more closely related to the poverty measure. The available ranges or buckets of each outcome, for which individual or household counts are provided, change across census years. Consequently we combine some buckets in certain years to maintain consistent ranges over time. However, to fully exploit the available distributional information we attempt to keep as many buckets as possible for each outcome.[9] Using our data, we are able to investigate changes in the density of individuals or households at 9 points in the distribution of household income relative to the poverty line, 13 points in the distribution of annual household income and 16 points in the distribution of monthly gross rent and house values.[10]

Note that one concern with this approach is that ranges of income or rent are specified in consistent nominal buckets over time because there is no straightforward way to create consistent ranges in real terms from the aggregated data. However, we are using a differencing approach where the EZ and control areas are both differenced over the same time span so changes in prices levels will be differenced out. Additionally, the poverty line is adjusted over time based on inflation so the ratio of household income to poverty line implicitly accounts for inflation. Finding similar results using the poverty measures as when we use household incomes suggests that our use of nominal buckets is not driving the results.

We are interested in estimating how EZ designation changes each of these distributions of individuals and households. To do so, however, requires the identification or construction of a relevant counterfactual distribution against which the observed changes in the EZ areas can be compared. Denoting the probability density function of outcome θ as $g(\theta)$, the change in the density of the distribution between 1990 and 2000 is defined as $\delta = g(\theta)_{2000} - g(\theta)_{1900}$. Letting δ_1 be the change in the distribution that would occur if an area receives EZ designation and δ_0 be the change that would occur without EZ designation, we are interested in identifying the treatment on the treated (TT) defined as

$$\mathrm{TT} = (\delta_1 | \mathrm{EZ} = 1) - (\delta_0 | \mathrm{EZ} = 1. \quad (1)$$

The primary identification problem is that while the first term is easily identifiable, it is simply the observed change in the probability density function for EZ areas, the second term is not. The second term is the counterfactual change in the probability density function that would occur in the areas that receive EZ designation if these areas did not receive EZ designation.

We will use the EC areas as a basis for constructing the counterfactual distributions, as has been done previously in the EZ literature, because these areas all applied and qualified for the program but were not selected. However as noted by previous researchers (e.g. Ham et al., 2011; Busso et al., 2013; Reynolds and Rohlin, 2014), while the EC areas all qualified for the EZ program, much of the EC areas were markedly better on a variety of observable characteristics than the EZ areas. Table 1 presents summary statistics for our main outcomes for the EZ and EC areas, as well as the differences in the outcomes between EZ and EC areas. In both 1980 and 1990, EZ areas had a higher poverty rate as well as lower annual household incomes, monthly gross rent and house values, all differences statistically significant from EC areas.

Similar to Busso et al. (2013), we construct our counterfactual by reweighting the EC areas to be more similar to the EZ areas in the years prior to EZ designation in 1994. We reweight using the propensity score $P(X)$, defined as the probability of being in the treatment group (EZ) conditional on some set observable characteristics X (Rosenbaum and Rubin, 1983).[11] For each outcome we consider, we estimate the propensity score as a function of lagged

[8] EZ and EC areas are designated by 1990 census geography so we convert 1980 and 2000 data to 1990 geographic equivalents using GIS maps to maintain consistent geography across the sample. Block group geography does not substantially change over time. The median 2000 census block group has approximately 95% of its area mapped to a 1990 census block group and there is a 0.86 correlation in block group household counts between 1990 and 2000.

[9] There are substantially fewer income buckets in the 1980 data compared to the 1990 and 2000 data. Combining the 1990 and 2000 income buckets to match the 1980 data would discard much of the information about distributional changes between 1990 and 2000 that we are interested in studying. Since we only use the 1980 distribution to control for trends, it does not need to have the same buckets as the 1990 and 2000 data. Therefore, we use the available 1980 distribution and then construct identical buckets between the 1990 and 2000 data.

[10] For household income we combine three income buckets into a single category for annual household income is $100 K or higher. We also combine together three buckets of house values into a single category of $300 K or higher. We do this because the impoverished nature of EZ and EC areas means that the density of households above these levels is very small in both 1990 and 2000. All results are robust to not making this adjustment and are available upon request.

[11] While reweighting or matching methods have been used in the prior literature, as discussed previously, other researchers have utilized an instrumental variable approach to account for the endogeneity of EZ designation. Pioneered by Hanson (2009) and subsequently used by Hanson and Rohlin (2011a, 2011b), this approach exploits differences in political characteristics of the applicant's federal representative and generally finds the EZ program to be less successful. See Hanson (2009) for further details.

Table 1
Selected summary statistics for EZ and EC areas.

	EZ		EC		Difference
	Mean	St. dev.	Mean	St. dev.	
1990					
Poverty rate	0.466	0.171	0.407	0.168	0.059***
Household income ($1000)	18.346	7.332	19.765	7.509	−1.419***
Gross rent ($1000)	0.346	0.101	0.378	0.124	−0.033***
House value ($1000)	45.530	44.468	64.051	45.619	−18.521***
1980					
Poverty rate	0.304	0.139	0.234	0.128	0.070***
Household income ($1000)	10.820	3.497	11.898	3.709	−1.078***
Gross rent ($1000)	0.185	0.049	0.197	0.051	−0.011***
House value ($1000)	21.312	12.870	30.932	17.008	−9.620***
N	838		3649		

Notes: (1) Asterisks denote whether the difference in means between the EZ and EC areas are statistically significant at the 10% (*), 5% (**) and 1% (***) levels.

1980 and 1990 measures of the empirical distributions. Specifically, for each outcome θ we use a logit to estimate

$$\text{EZ} = f(g(\theta)_{1900}, g(theta)_{1980}, \tag{2}$$

where $g(\theta)_{1990}$ and $g(\theta)_{1980}$ are the empirical probability density functions in the block group in 1990 and 1980, respectively. The propensity score $\hat{P}(g(\cdot))$ is the probability of a block group being in an EZ predicted by Eq. (2). Thus, when estimating the propensity score we are controlling for the pre-designation distribution of θ as well as any trends in the distribution prior to EZ designation.[12]

The empirical probability density functions are represented by a set of variables measuring the fraction of total individuals or households within the available ranges of each distribution as described above. While block groups are constructed to be similar in population size, some variation in population does exist. Consequently, we estimate Eq. (2) and calculate propensity scores separately by quintiles of block groups based on population. In practice this has little impact on the estimates and similar procedures, for example including indicators for the size of the block group in Eq. (2), produce similar estimates.

Using the propensity score $\hat{P}(g(\cdot))$, we construct the counterfactual changes by reweighting the EC areas by weight $\omega = \frac{\hat{P}}{1-\hat{P}}$.[13] Implicitly, this procedure upweights those parts of the EC areas that are most similar in their 1980 and 1990 distributions of θ to EZ areas while downweighting those parts of the EC areas which are most unlike the EZ area. This particular use of the propensity score is often less biased and more efficient than other matching methods when there is sufficient overlap of the propensity scores in the treated and control sample (for a discussion see Busso et al. (2014)). However, we later demonstrate that all of our substantive results are unchanged if we create counterfactuals using other algorithms based on the propensity score.

Our main identifying assumption for the reweighting procedure is

$$(\delta_0|(X), \text{EZ} = 1) = (\delta_0|P(X), \text{EZ} = 0, \tag{3}$$

meaning that conditional on the propensity score, there is no difference in the change in the distribution of θ under non-treatment across EZ designation. This assumption simply states that, conditional upon the propensity score, the observed change in the distribution of θ in EC areas serves as the appropriate counterfactual to estimate Eq. (1). We will demonstrate that the reweighting procedure produces a control group with 1980 and 1990 empirical distributions that are nearly identical to the corresponding distributions in the EZ areas. Thus, our control experiences the same changes as the EZ areas throughout the distribution of each outcome prior to EZ designation. Thus, any bias in our distribution estimates must be due to our control areas systematically beginning to trend differently after 1990, despite having trended similarly throughout the distribution prior to 1990. While we acknowledge that this is possible, we feel that it is unlikely given our rich data on the pre-designation distributions for each outcome we consider.[14]

4. Results

4.1. Average effects

To understand the importance of studying distributional changes we first use our matching procedure and block group data to estimate the average effects on our primary outcomes, similar to much of the prior literature. For each of our outcomes, we first estimated the propensity score on the 1990 and 1980 levels of the outcome and then used the reweighting procedure to construct the counterfactual change from the EC areas. The reweighting procedure produces counterfactual 1980 and 1990 levels for each of the four outcomes considered that are much closer to those of the EZ areas and any small differences are not statistically significant (results available upon request).

Table 2 presents the estimated effects of EZ designation on the poverty rate and real mean and median annual household income, monthly gross rent and house values using our matching methodology. For each outcome, the observed change in EZ areas from 1990 to 2000 is presented in column 1, the estimated counterfactual change constructed from the EC areas using the reweighting procedure is presented in column 2, and column 3 presents the TT, the difference between the observed and counterfactual change as defined by Eq. (1). Asterisks denote statistical significance based on the empirical confidence intervals produced by bootstrapping the estimates using 1000 replications; 95% confidence intervals are presented in square brackets below point estimates.[15]

We begin by considering the poverty rate in EZ areas. The result in the first row suggests that the proportion of the population in EZ areas below the poverty line fell by 7.8 percentage points from 1990 to 2000. However, the estimated counterfactual suggests that the poverty rate would have fallen by 6.7 percentage points in the absence of the EZ program. Therefore, the effect of receiving the EZ program for the designated areas is a reduction in the poverty rate of 1.0 percentage point, an effect that is neither economically nor statistically significant.[16] Thus, we find little evidence that the EZ program reduced average poverty.

[12] One potential concern is that the use of lagged outcomes as controls when estimating the propensity score could result in division bias (Borjas, 1980) because the 1990 measures are included in both the controls and in the outcome. However, in Section 4.2.2 we argue that this is not likely to drive our distributional estimates and demonstrate that all results are robust to imputing distributions using nearby block groups.

[13] In practice, we normalize ω to sum to one.

[14] Eq. (3) is less strong than the assumption required to use the observed changes in EC areas as counterfactuals, which has been done frequently in the literature. The corresponding identifying assumption to use the observed changes in EC areas as counterfactuals could be expressed as $(\delta_0|\text{EZ} = 1) = (\delta_0|\text{EZ} = 0$, meaning that regardless of any differences in the observed characteristics of the areas, the distribution of θ would change the same over time regardless of EZ designation. This is only likely to hold if there is little difference between EZ and EC areas. Our procedure attempts to account for the fact that significant differences do exist between those areas that did and did not receive EZ designation.

[15] Throughout the paper, we use a block-bootstrap procedure where we sample from within the EZ and the EC areas so that each sample has the same proportion of EZ and EC block groups that we observe in the block group data.

[16] In contrast, the observed change in the poverty rate in EC areas is only −0.044. Using this as the counterfactual change for the EZ areas would produce a larger estimated effect of the EZ program on the poverty rate.

Table 2
Effects of EZ designation on the poverty rate and mean and median annual household income, monthly gross rent and house values.

Outcome	Change in EZ (1)	Counterfactual change (2)	Difference (ATT) (3)
Poverty rate	−0.078***	−0.067***	−0.010
	[−0.091, −0.063]	[−0.077, −0.058]	[−0.022, 0.003]
Average			
Annual household income ($1000)	13.599***	11.555***	2.044***
	[12.686, 14.577]	[11.063, 12.115]	[1.021, 3.092]
Monthly gross rent ($1000)	0.139***	0.128***	0.010*
	[0.130, 0.148]	[0.126, 0.135]	[−0.001, 0.022]
House value ($1000)	50.042***	22.912***	27.13***
	[42.253, 59.040]	[21.366, 24.444]	[19.545, 36.152]
Median			
Annual household income ($1000)	8.121***	7.870***	0.251
	[7.406, 8.792]	[7.447, 8.337]	[−0.570, 1.077]
Monthly gross rent ($1000)	0.132***	0.132***	0.000
	[0.121, 0.142]	[0.125, 0.139]	[−0.012, 0.014]
House value ($1000)	42.962***	21.880***	21.082***
	[35.843, 50.899]	[19.926, 23.933]	[13.428, 29.456]

Notes: (1) Column (1) presents the observed change in EZ areas from 1990 to 2000, column (2) presents the counterfactual change constructed by propensity score reweighing of the EC areas. Column (3) presents the difference between the observed and counterfactual change, the average treatment on the treated (ATT). Each row represents a separate estimation procedure. (2) 95% confidence intervals are presented in square brackets below the point estimates. The confidence intervals are based on the empirical distributions following a bootstrap procedure with 1000 replications. Asterisks denote statistical significance at the 10% (*), 5% (**) and 1% (***) levels.

Somewhat in contrast, however, the results in the middle panel suggest that the federal EZ program on average increased annual household income by approximately $2044. This is roughly an 11% increase relative to the baseline household income level in EZ areas in 1990. Additionally, we find that EZ designation increased gross rents in these areas by approximately $33 a month from 1990 to 2000 and increased house values by $27,130. These results are broadly consistent with the previous literature which has found little increase in gross rents (Busso et al., 2013) but large increases in house values (e.g. Hanson, 2009; Krupka and Noonan, 2009; Busso et al., 2013).[17]

The results in Table 2 suggest that there may be potentially interesting changes in the distributions of these measures. For example, while the poverty rate does not decline, household income appears to increase on average which would be consistent with an increase in household incomes among households above the poverty line. Additional evidence that analysis of mean effects are missing potentially important changes in the underlying distributions can be found in the bottom panel of Table 2 where instead of using the mean values of household income, gross rent and house values in each block group, which can be skewed by extreme values, we instead look at median values. The results in column 3 suggest that EZ designation had less effect on both median household income and median gross rent than was estimated using average values. For neither median household income nor median gross rent are the estimated effects large or statistically significant. Additionally, the estimated impact on median house values is smaller than the estimated mean effect, although the effect is still large. Overall, these results suggest that estimates of EZ designation on average household income and gross rents could be influenced by large increases at the top of the distributions, and thus analysis of the entire distribution would be valuable in determining who is benefiting from this policy.

4.2. Distributional effects

4.2.1. Household income relative to the poverty level

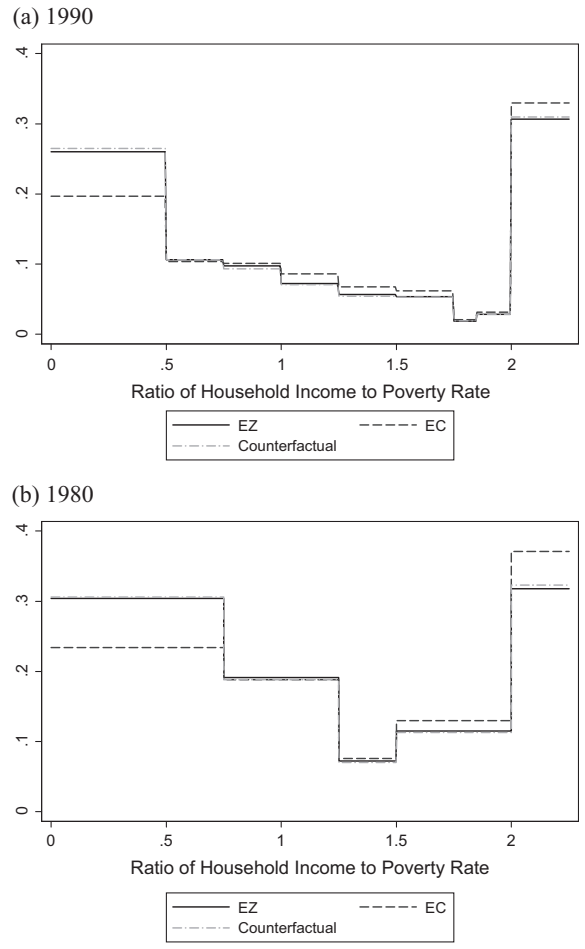
We begin our exploration of the potential distributional effects of the federal EZ program by examining how EZ designation changes the density of households across the distribution of household income relative to the poverty line. We first estimate the propensity score using measures of the 1980 and 1990 densities of the population in specified ranges of the distribution of household income relative to the poverty line. We then use the estimated propensity score to construct the counterfactual changes in the density between 1990 and 2000 across the distribution.

We focus the discussion using a graphical presentation of the results because it more clearly illustrates the complex changes in the densities of the relevant distributions. For example, Fig. 1 presents a graphical investigation of the balancing property of our reweighting procedure for the 1990 and 1980 densities of households across the distribution of household income relative to the poverty level. Consistent with the summary statistics presented in Table 1 about differences between EZ and EC areas in the overall poverty rate, both the 1990 and 1980 poverty distribution is more highly skewed in the EZ areas with a larger density of households at lower poverty levels compared to the EC areas. However, the counterfactual distributions in each year, constructed by reweighting the EC areas, closely match the distribution in EZ areas.[18] Thus, the matching method produces a counterfactual distribution that closely follows the pre-EZ designation trends in the distribution of households relative to the poverty line.

The effect of EZ designation on the density of households relative to the poverty line is depicted in Fig. 2, with the poorest income bucket being 0–0.5 of the poverty line and the richest income bucket being 2 or more times the poverty line. The first row of figures presents the observed distributions in 1990 and 2000 for EZ areas (2a) and the equivalent counterfactual distributions (2b). Both demonstrate a decrease in the density of households in extreme poverty and an increase in the density of households with incomes at least twice the poverty line. These

[17] Our estimated mean results are similar to Busso et al. (2013). In log terms, we find a 0.021 (0.040) increase in mean rent, a 0.259*** (0.040) increase in mean house value, and 0.059*** (0.017) for mean household income. Busso et al. (2013) find, depending on the specification, approximately a 0.02 increase in log rent, a 0.30 increase in log value, and a 0.05 increase in weekly wage income.

[18] Estimates not shown here but available upon request show no statistically significant differences between the poverty rate distributions in 1980 and 1990 between the EZ and reweighted EC areas.



Notes:
(1) The counterfactual distribution is produced by propensity score reweighting of the EC distribution.

Fig. 1. Densities of population across the distribution of household income relative to the poverty level in 1990 and 1980 in EZ and EC areas.

changes in the densities along the distributions are displayed, with 95% confidence intervals, in the second row for EZ areas and the counterfactual, respectively. Fig. 2c corresponds to $\delta_1|\text{EZ}=1$ and Fig. 2d corresponds to the estimated counterfactual $\delta_0|\text{EZ}=1$ needed to calculate the treatment on the treated in Eq. (1).

These figures, and the corresponding point estimates in columns 1 and 2 of Table 3, suggest that the EZ areas experienced a 5.1% decrease in the density of individuals with household income less than half the poverty line and smaller declines in the densities of individuals with household income at least half, but still below, the poverty line. While there is an increase in the density of individuals slightly above the poverty line, potentially because of individuals moving out of poverty, most of the change in the distribution appears to be occurring at the top of the distribution with a 5.8% increase in the density of individuals with household income twice the poverty line.

While the counterfactual shows a similar pattern in terms of the location and direction of changes in the densities of the distribution (Fig. 2d or column 2 of Table 3), the magnitudes are not identical. Designation of EZ status in EZ areas, the treatment on the treated (TT) or difference between the observed and counterfactual changes, appears to polarize the poverty distribution. Fig. 2e shows that relative to the counterfactual, the density of households in EZ areas increases at both ends of the distribution while the density decreases in the middle of the distribution. The corresponding point estimates in column 3 of Table 3 suggest that the density of individuals in extreme poverty, with household incomes less than half of the poverty line, increases by 1.1%. However, there is also a 1.9% increase in the density of individuals with household income at least twice the poverty line.

As we demonstrated in Table 2, these changes at both the top and the bottom of the poverty distribution are not observed when estimating the effect of EZ designation on the overall poverty rate.

4.2.2. Robustness checks

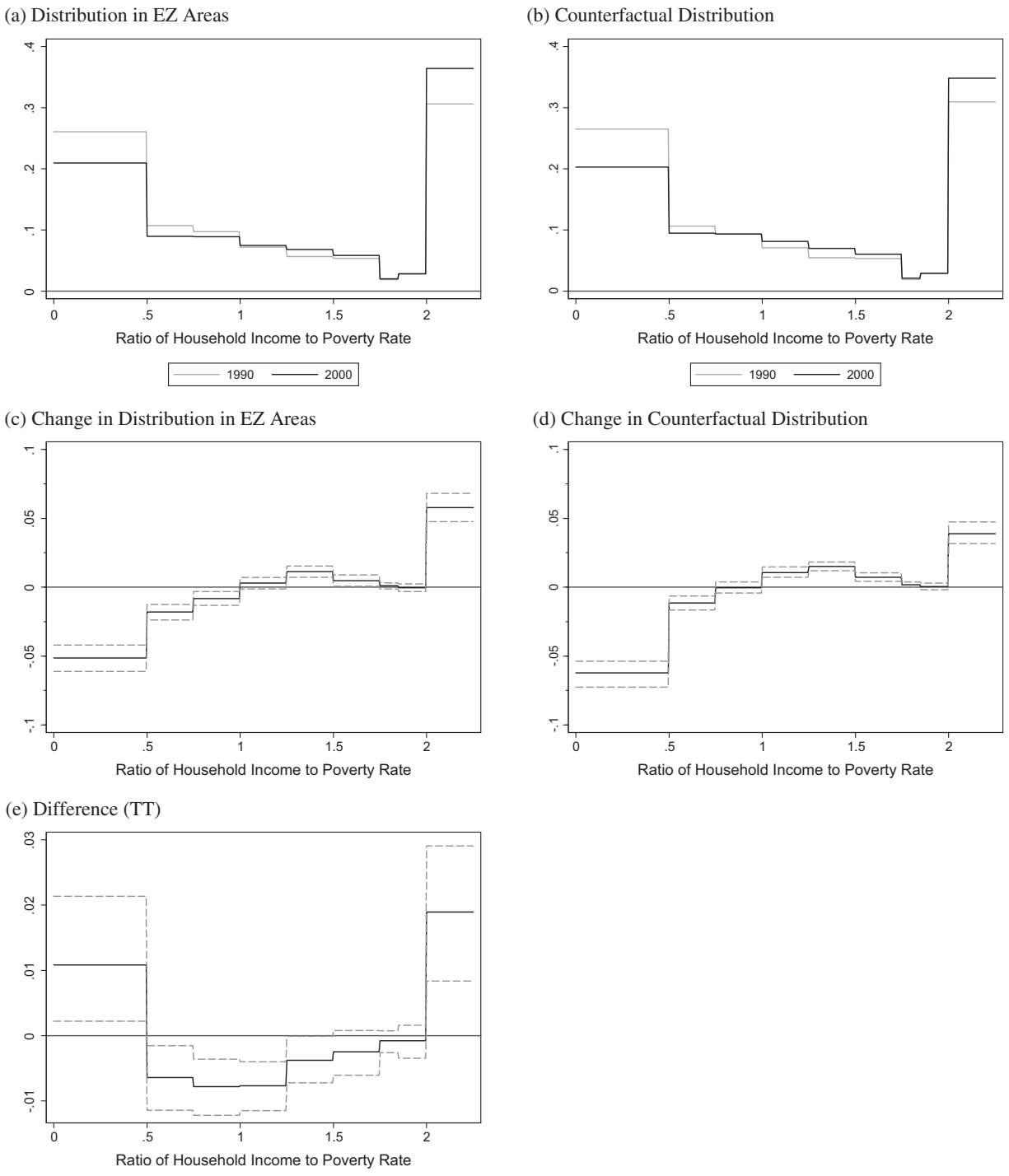
The results we present in the previous section are robust to a number of methodological choices and potential issues. First, the polarization pattern is robust to including population quintile indicators when estimating the propensity score in our reweighting procedure (column 2 of Appendix Table A1) or using other matching algorithms based on the propensity score although, as discussed, other algorithms are less precisely estimated (see column 3 of Appendix Table A1).

As mentioned previously, researchers using difference-in-difference methods must be careful for division bias (Borjas, 1980) due to mis-measurement in lagged variables when those variables are in both the outcome (in our case, the change from 1990 to 2000 in some outcome) and in the controls (when we match on 1980 and 1990 values). While this is certainly possible in many contexts, measurement error in the 1990 variables is unlikely to drive our estimated distributional changes because it would depend on systematic mis-measurement in specific parts of each distribution in that year across EZ and EC areas. Furthermore, while measurement error in the income of each household could be problematic when estimating mean effects, such measurement may have little effect on the counts of households in each range of household income because it does not necessarily shift households to the "wrong" income bucket. However, we test for such problems by imputing the distribution in each block group using a weighted average of the distributions in nearby block groups (where weights are Gaussian across distance within 1 mile and zero beyond 1 mile). While the precision of the estimates decreases (column 4 of Appendix Table A1), the overall pattern of bimodal changes remains.

Additionally, some of the prior EZ literature has argued for using the census tracts that border EZ areas to construct counterfactuals. However, Hanson and Rohlin (2013) find evidence that EZ areas attracted businesses from nearby census tracts, suggesting that EZ bordering tracts are themselves treated through negative spillovers. In fact, one interpretation of the evidence we present subsequently is that the bimodal distributional changes we observe are due to movement of households, likely within the city. Thus, we prefer our method of constructing counterfactuals from the EC areas. However, we can apply our methodology, comparing the EZ-bordering areas to the EC-bordering areas, to investigate whether our EZ results are due simply to local changes that are not contained only within the EZ areas.

The results, presented in column 5 of Appendix Table A1, indicate that the EZ-bordering areas did not experience a bimodal change in the distribution; instead they appear to be generally decreasing in income, with a 5.7 percentage point increase in households at less than half the poverty line and a 4.7 percentage point decrease in households at twice the poverty line. Thus, the relative increase in households greater than 200% of the poverty that we find in EZ areas do not appear to be due to local effects within and near the EZ areas, since the areas surrounding EZs actually became more impoverished relative to the areas surrounding EC areas.

Overall, the bimodal results we find are robust to several methodological concerns. Importantly, however, the results in



Notes:
(1) Counterfactual distributions are calculated from EC areas using propensity score reweighting.
(2) Dashed lines in figures c) – e) represent 95% confident intervals calculated from a bootstrapping procedure with 1000 replications. Corresponding point estimates and confidence intervals are presented in columns 1 through 3 of Table 3, respectively.

Fig. 2. Effect of EZ designation on the density of population across the distribution of household income relative to the poverty level.

Table 3 are also broadly consistent with changes in a number of additional distributions, as we subsequently document.

4.2.3. Household income

The possibility that the EZ program has effects at the very top of the income distribution can be investigated with the same methodology using data on the distribution of household income. Importantly, the household income data has more buckets, particularly at incomes higher than twice the poverty line (approximately $35,000 in 2000), allowing for a more detailed exploration of where in the income distribution changes are occurring. Fig. 3a–c demonstrate that EZ areas see a large decrease in the density of households with less than $10,000 in annual income and a comparative increase in the density of households with income

Table 3
Effect of EZ designation on the density of population across the distribution of household income relative to the poverty level.

Ratio of household income to poverty level	Change in EZ (1)	Counterfactual change (2)	Difference (TT) (3)
0–0.49	−0.051***	−0.062***	0.011**
	[−0.063, −0.040]	[−0.075, −0.052]	[0.001, 0.024]
0.50–0.74	−0.018***	−0.012***	−0.006**
	[−0.025, −0.012]	[−0.018, −0.005]	[−0.012, −0.001]
0.75–0.99	−0.008***	−0.000	−0.008***
	[−0.014, −0.002]	[−0.005, 0.005]	[−0.013, −0.003]
1.00–1.24	0.003	0.011***	−0.008***
	[−0.002, 0.008]	[0.007, 0.015]	[−0.012, −0.003]
1.25–1.49	0.011***	0.015***	−0.004*
	[0.007, 0.016]	[0.011, 0.019]	[−0.008, 0.001]
1.50–1.74	0.005*	0.007***	−0.002
	[−0.000, 0.009]	[0.004, 0.011]	[−0.007, 0.002]
1.75–1.84	0.001	0.002	−0.001
	[−0.002, 0.004]	[−0.000, 0.004]	[−0.003, 0.001]
1.85–1.99	−0.000	0.000	−0.001
	[−0.004, 0.003]	[−0.002, 0.003]	[−0.004, 0.002]
2.00+	0.058***	0.039***	0.019***
	[0.045, 0.070]	[0.030, 0.049]	[0.007, 0.030]

Notes: (1) Column (1) presents the observed change in EZ areas from 1990 to 2000, column (2) presents the counterfactual change constructed by propensity score reweighing of the EC areas. Column (3) presents the difference between the observed and counterfactual change, the treatment on the treated (TT). (2) 95% confidence intervals are presented in square brackets below the point estimates. The confidence intervals are based on the empirical distributions following a bootstrap procedure with 1000 replications. Asterisks denote statistical significance at the 10% (*), 5% (**) and 1% (***) levels.

above $50,000. As discussed previously, to maintain consistent buckets with which to calculate densities, these income ranges are specified in nominal terms so much of this increase in income is due simply to inflation, which is reflected in a similar pattern being seen in the counterfactual distributions in Fig. 3b and d. However, since both areas are changing over the same period of time, the difference between the observed changes in EZ areas and the counterfactual accounts for national price trends.

The results in Fig. 3e depict a similar polarization of the distribution as seen in the poverty distribution (numeric estimates available upon request). EZ areas experience a 1.4% increase in the density of households with annual income below $10,000 compared to the counterfactual, consistent with the increase in extreme poverty previously seen in Fig. 2. Additionally there is a small positive, but not always statistically significant, increase in the density of households with income above $50,000. However, there is a larger statistically significant increase of 0.8% in the density of households with incomes above $100,000. These results suggest that the increase in households incomes is occurring well beyond twice the poverty line, which seems unlikely to be caused by either increased wages or employment associated with the $3000 wage credit. Instead, Fig. 3 provides further evidence that prior estimates of increased average incomes and wages may be skewed by increases at the top of the income distribution, consistent with the large mean but not median effects on household income that we demonstrated in Table 2.

4.2.4. Gross rent and house values

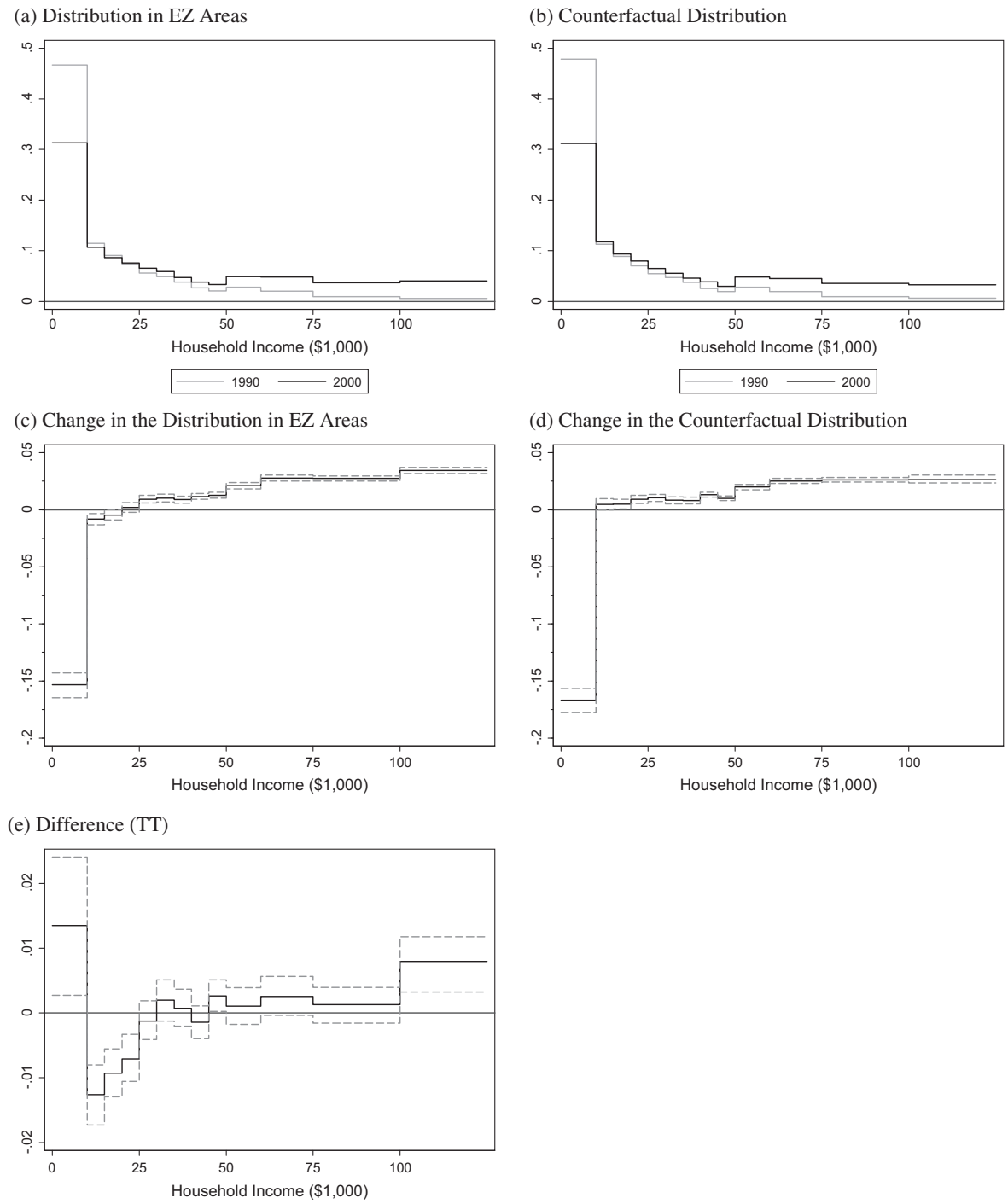
In addition to the poverty rate and income, the previous literature on the EZ program has frequently investigated house values and gross rents because these measures can be interpreted as proxying for local amenities and the overall attractiveness of the areas. Similarly, we can use our methodology to investigate how the EZ program affected the distributions of rental rates and house values in EZ areas. For simplicity, we only present the difference between the observed change and the counterfactual change (TT) for each outcome in Fig. 2 but full estimates and figures are available upon request.

Compared to the household income estimates, there is less evidence of a polarizing effect of EZ designation on housing costs. There appears to be an increase in the density of households paying monthly gross rents above $600 and a decrease in the density of households paying gross rents between $350 and $550. This does suggest an increase in rental rates, but an increase that occurs in the top half of the distribution consistent with an increase in the rental rates of mid-range rental units only. This movement in the middle of the distribution may explain why prior estimates have generally found only modest, or no, positive effects on average rental rates.

In contrast, there are appears to be a large increase in house values starting at $100,000 and extending above $300,000. This increase at levels several times the average or median 1990 house value in EZ areas partly explains why the prior literature has consistently found increases in mean house values. Overall, this evidence suggests that the attractiveness of the EZ areas to households increased due to the program, as reflected in higher costs of living, but the fact that the effect is larger for owner-occupied housing and the effects on house values extends so far up the distribution makes one question what is driving these changes.

5. Discussion

Our results suggest complex effects of EZ designation on the *distribution* of income and poverty in treated areas that are missed when analyzing mean or median outcomes. We find a polarizing effect of EZ designation on the distribution of individuals relative to the poverty line and the corresponding overall distribution of household income. How should we interpret this polarized effect? Despite some evidence of positive effects in the prior literature using mean estimates (e.g. Busso et al., 2013; Ham et al., 2011; HUD, 2001; Krupka and Noonan, 2009; Reynolds and Rohlin, 2014), we find no evidence that the federal EZ program lifted households out of poverty. First, while the overall density of the very poor did decrease in EZ areas, there was less of a decrease compared to the counterfactual areas. Second, even if the EZ



Notes:
(1) Counterfactual distributions are calculated from EC areas using propensity score reweighting.
(2) Dashed lines in figures c) – e) represent 95% confident intervals calculated from a bootstrapping procedure with 1000 replic ations. Corresponding point estimates and confidence intervals are available upon request.

Fig. 3. Effect of EZ designation on the density of households across the distribution of household income.

program did increase wages for the poor, the increased density of high-income households we observe is likely not due to a movement up the income distribution of those individuals living in the EZ areas prior to 1994. It seems implausible to believe that a $3000 wage credit and the Social Service Block Grants could move households from below the poverty line to more than twice the poverty line, or cause households to earn more than $100,000 in 2000 when the average pre-designation income in EZ areas was approximately $25,000 (in real 2000$).

Instead, our results are more consistent with a mobility response leading to a change in the distribution of households in EZ areas. In particular, it may be that EZ designation makes the areas more attractive to higher income households who move into the area causing gentrification of the zone, which would support

Table 4
Effect of EZ designation on the density of population across levels of educational attainment and by gross rent as a percent of household income.

	Change in EZ (1)	Counterfactual change (2)	Difference (TT) (3)
Highest level of educational attainment			
Less than high school	−0.097***	−0.064***	−0.033**
	[−0.108, −0.087]	[−0.073, −0.055]	[−0.045, −0.020]
High school degree	0.011***	0.011***	0.000
	[0.003, 0.018]	[0.004, 0.17]	[−0.007, 0.008]
Some college	0.048***	0.032***	0.016***
	[−0.041, 0.055]	[0.026, 0.038]	[0.009, 0.023]
Bachelor's degree	0.023***	0.015***	0.008**
	[0.018, 0.027]	[0.010, 0.021]	[0.001, 0.014]
Graduate degree	0.016***	0.007***	0.010***
	[0.012, 0.020]	[0.004, 0.009]	[0.005, 0.014]
Gross rent as a percentage of household income			
0–0.20	0.063***	0.032***	0.031***
	[0.051, 0.072]	[0.023, 0.043]	[0.016, 0.042]
0.20–0.25	0.006	0.007*	−0.001
	[−0.002, 0.013]	[−0.001, 0.014]	[−0.009, 0.007]
0.25–.30	−0.002	−0.000	−0.002
	[−0.009, 0.006]	[−0.008, 0.008]	[−0.011, 0.006]
0.30–0.35	−0.003	−0.000	−0.003
	[−0.009, 0.003]	[−0.006, 0.006]	[−0.010, 0.002]
0.35+	−0.063***	−0.039***	−0.025***
	[−0.075, −0.050]	[−0.051, −0.025]	[−0.039, −0.009]

Notes: (1) Column (1) presents the observed change in EZ areas from 1990 to 2000, column (2) presents the counterfactual change constructed by propensity score reweighing of the EC areas. Column (3) presents the difference between the observed and counterfactual change, the treatment on the treated (TT). (2) 95% confidence intervals are presented in square brackets below the point estimates. The confidence intervals are based on the empirical distributions following a bootstrap procedure with 1000 replications. Asterisks denote statistical significance at the 10% (*), 5% (**) and 1% (***) levels. (3) Educational attainment is measured for the population aged 25 or older. Gross rent as a percentage of household income is measured for renters.

the evidence of increased house values at the upper portion of the distribution.[19] Additionally, the fact that EZ areas appear to see an increase in the proportion of individuals in severe poverty could suggest that the program has either also attracted very poor households to the area or has failed to help the very poor in previously living in the EZ area. Regardless, any potential positive benefits of the program do not appear to accrue equally to all individuals within EZ areas, a result that is not clear from mean estimates. Furthermore, our results raise questions about whether location-based policies can be used successfully to target a particular subpopulation, such as the local poor, when the effect of the program is to encourage workers and firms to relocate across space. In the remainder of this section, we provide additional evidence that such a relocation of households appears to occur following EZ designation.

5.1. Educational attainment and housing expenditures

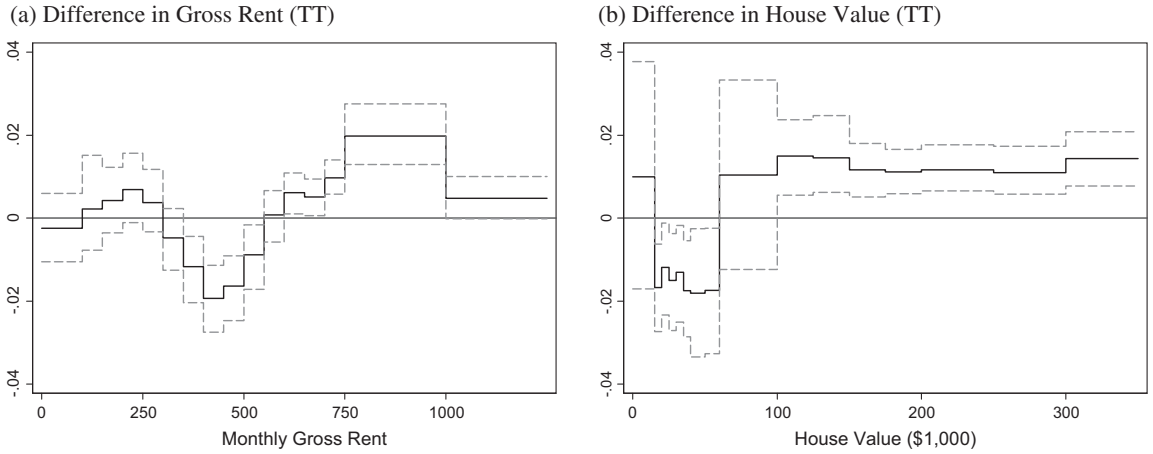
To further investigate whether household migration explains some of the changes in the distributions above, and by extension some of the prior mean effects estimated in the literature, we examine both changes in educational attainment and gross rent relative to household income. Table 4 presents the estimated impact of EZ designation on these two distributions of the population. Consistent with their impoverished nature, EZ areas had very low rates of educational attainment in 1990 prior to EZ designation with approximately 55% of individuals aged 25 or older having not completed high school and another 25% only completing high school. However, the top panel demonstrates that EZ-designation is associated with an increase in educational attainment with a 3.3% decrease in the density of individuals aged 25 or higher with less than a high school diploma. Furthermore, this reduction in the density is due to an increase in the density of individuals with at least some college experience or higher, including a 0.8% increase in the proportion of individuals with a bachelor's degree and a 1.0% increase in the proportion of individuals with a graduate degree. This is likely due to an in-migration of highly educated individuals as it seems implausible that a $3000 wage credit and any corresponding employment increase among the EZ population would have caused large increases in educational attainment within 6 years, particularly at the level of bachelor or graduate degree holders.

The results in the previous section suggested that both household incomes and housing costs increased. However, the lower panel of Table 4 suggests that the increase in household incomes in the EZ areas is outpacing the increase in housing costs, at least for renters, as the proportion of households paying comparatively little of their annual income in the form of rent increases while the proportion of households committing a larger portion of their annual income to rent decreases. Given the increase in rental rates at the middle to the top of the distribution observed in Fig. 4, these results would be consistent with higher income individuals moving to a relatively low cost area.

5.2. Spatial distribution

One interpretation of our results is that they could be due to the movement of individuals and households into the EZ areas but they could also be due to spatial variation in the concentration of effects within EZ areas. To further explore these possibilities, we investigate whether the changes in poverty varies within each city based on the initial level of poverty in 1990. Importantly, while EZ areas

[19] There is some prior evidence about in-migration of residents to EZ areas. Busso et al. (2013) find a 0.02% change in households, a 0.06% change in population in EZ areas and a small decrease in the 5-year housing turnover rate, although none of the estimates are statistically significant. Similar to evidence we will present, they do find that the proportion of the population with a college degree increases by 0.02. We have also found evidence of changes in racial composition with a change in white residents in EZ areas of −0.022*** (0.006), change in Asian residents of −0.003** (0.001), and change of black residents of 0.033*** (0.008), relative to EC areas. The racial changes suggest a mobility response, but it is unclear whether it is due to in-migration of poor black households, consistent with the increase at the bottom of the income distribution, or in-migration of high-income black households, consistent with the increase at the top of the distribution, or a combination of both.



Notes:
(1) Counterfactual distributions are calculated from EC areas using propensity score reweighting.
(2) Dashed lines in figures a) – b) represent 95% confident intervals calculated from a bootstrapping procedure with 1000 replications. Corresponding point estimates and confidence intervals are available upon request.

Fig. 4. Effect of EZ designation on the density of households across the distribution of monthly gross rent and house values.

are defined according to census tracts and the area overall had to meet the minimum criteria to qualify for the EZ program, each individual census tract within the EZ area did not have to meet the criteria. Thus, there is variation within each EZ area in terms of the poverty level.[20] The question is whether there is a different effect of EZ designation in the portion of each EZ area that is initially worse (ie. has higher initial poverty) or better (ie. has lower initial poverty) in 1990 prior to EZ designation.

To investigate this question, we repeat our estimation procedure separately within terciles of the 1990 poverty rate. We split the sample in two ways: first, we divide tracts across the EZ areas and second, we divide tracts within each EZ area. The former allows us to investigate the effects different ranges of initial poverty across the entire EZ sample while the latter provides information about how different parts of each EZ area responded to the program within each respective city. In each case, we limit the EZ tracts to the relevant tercile and then construct counterfactuals using the EC areas reweighted to have similar 1980 and 1990 poverty distributions.

Fig. 5 presents the estimated effects of EZ designation on the density of households relative to the poverty line, first by terciles across EZ areas in column (i) and then by terciles within EZ areas in column (iii). The graphs reveal several findings. First, it appears as if the increase in higher income households is concentrated in the tracts which were least poor in 1990, either across cities or within cities. However, within cities there does appear to be a small increase in higher-income households in the poorest tracts as well. In contrast, there is not much change in the distribution of income for the middle tercile of tracts. However, the increase in the lower-income households appears to be concentrated among the tracts which were poorest in 1990, particularly across cities.

These results suggest that the EZ program has effects not just across the distribution of poverty but furthermore across the spatial distribution of EZ areas. The results suggest that the EZ program appeared to have a more positive effect on the distribution of poverty among those portions of the EZ areas that were less impoverished initially, essentially more marginal for the EZ areas. This would seem to also be consistent with the EZ program causing a mobility response, in this case of high-income households moving into the EZ areas and disproportionately choosing to locate in the "best" part of these otherwise impoverished areas.

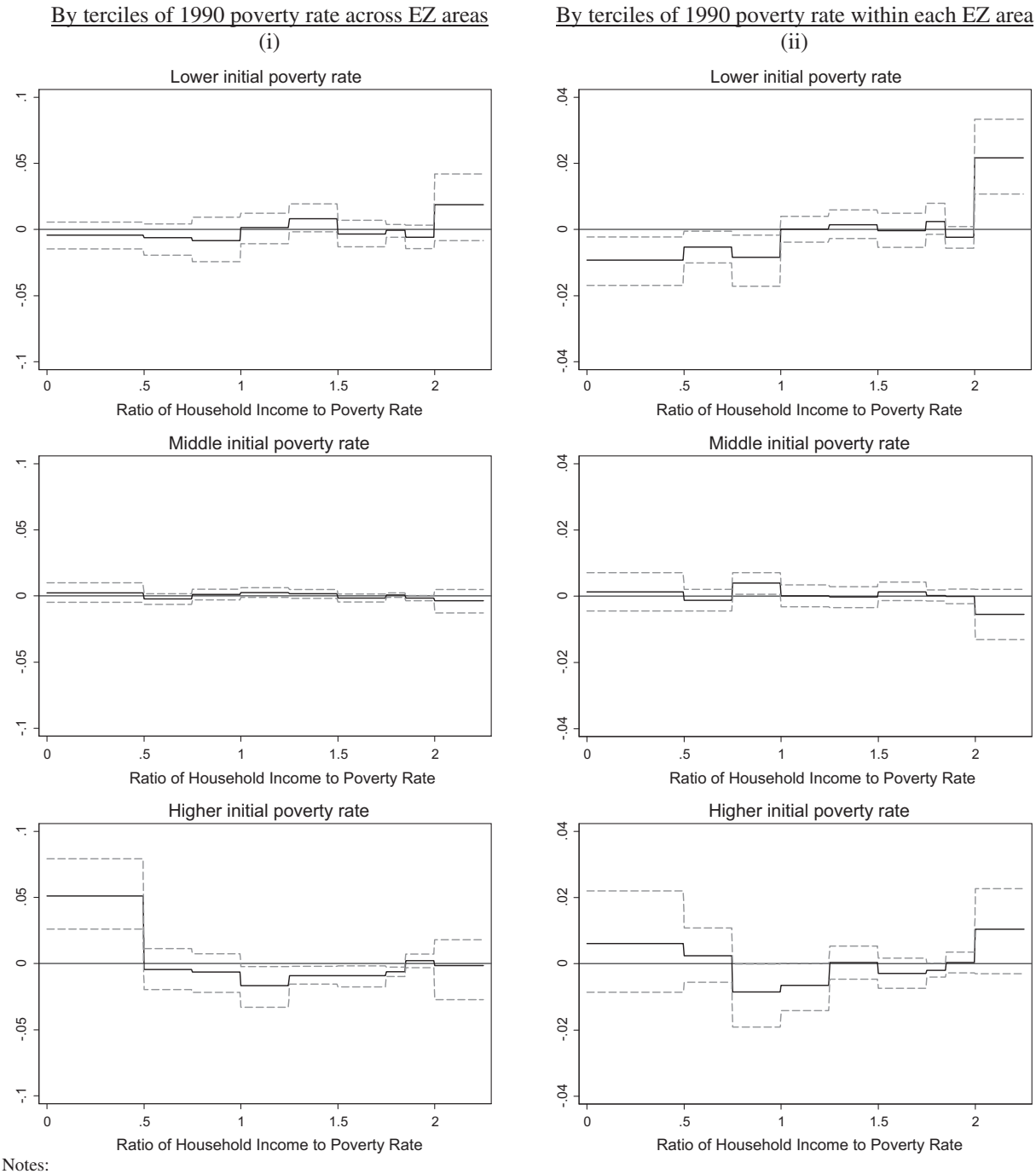
5.3. Lower-income households

While the evidence seems to suggest an in-migration of high income households, there remains a question of why there also appears to be a relative increase in very poor households as well. As mentioned previously in this section, it is possible that the EZ program just did not help prevent households from falling into poverty. The evidence from the bordering areas analysis in Section 4.2.2 suggest that the local area around EZ areas became more impoverished compared to the areas surrounding EC areas. Thus, the increase in very poor households may simply reflect this overall trend toward more poverty in the local area that the program failed to halt, despite appearing to attract high-income households to parts of the EZ areas.

A second possibility is that the mobility response extends beyond high income households. On the one hand, it is possible that the EZ program attracted more very poor households to the area, for example because the Social Service Block Grants were used to purchase amenities making the EZ areas relatively more attractive. It is also possible that the in-migration of high income households led to an exit of households in the middle of the distribution, for example because of higher rents. If this effect is large enough, it would hollow out the "middle" of the distribution leading to a relative increase of very poor households at the bottom of the distribution, because mechanically the changes throughout the distribution sum to zero.

Unfortunately, it is difficult to assess these types of mobility responses without individual household data. The possibility of higher-income households crowding out the middle of the distribution could be consistent with the concentrated movement of very high income individuals into the "best" part of the EZ areas. Additionally, a relative increase in very poor households relative to "middle" households could be augmented by the relative supply of low-cost housing. The large increase in very poor households across cities in column (i) of Fig. 5 is driven largely by the EZ areas in New York and Atlanta (results available upon request). New York had a very high concentration of public housing within the

[20] For example, the average tract-level poverty rate varies from 0.418 to 0.560 across EZ areas while the inter-quartile range within each area varies from 0.113 to 0.248 across EZ areas.



Notes:
(1) Dashed lines represent 95% confident intervals calculated from a bootstrapping procedure with 1000 replications.

Fig. 5. Change in the density of households across the distribution of household income relative to the poverty level by initial poverty rate in 1990.

EZ area while Atlanta began demolishing public housing projects just outside of the EZ area prior to 1996 Olympics, which increased the relative supply of public housing inside the EZ area. The relatively high supply of subsidized housing could have kept very poor households in the area, despite the increase in very high income households in parts of the EZ areas.

Unfortunately, it is difficult to separately test these various possibilities with currently available data. However, the increase in the relative proportion of very low income households following the EZ programs is a potential area of future research. Regardless of the mechanism, however, there is no indication that the EZ program helped raise households out of poverty.

6. Conclusion

The focus of this paper is to demonstrate the importance of analyzing distributional effects along with mean changes when investigating location-based policies such as the federal EZ program. Mean estimates can be difficult to interpret or can lead to inaccurate policy conclusions because the policies could have impacts in different parts of the distribution of the economic outcomes typically used to evaluate the program, such as household income, that are not readily revealed in mean estimates. Furthermore, these policies could reallocate households and firms across space which further complicates the interpretation of mean estimates.

Moving away from mean estimates, we find a disparate effect of the federal EZ program across different parts of the income distribution. Specifically, compared to similar areas that also applied to the EZ program, we find that there is an increase in the number of households that make less than half the poverty rate as well as an increase households with more than twice the poverty rate. Furthermore, our results are consistent with high-income households moving into the EZ areas. We demonstrate that the increase in households happens above $100,000 in household income and that there is also a significant increase in highly educated individuals. Additionally, we find that not all areas experience the growth of higher-income households with those EZ areas that were relatively well-off receiving the majority of the growth.

Our results may be unique to the federal program; however, given that other types of economic development programs have similar potential for redistribution, studying the distributional effect of these programs should be an area of future research. Our results highlight one of the benefits of studying distributional effects which is that distributional analysis can provide a more complete picture as to what is causing these mean changes. All of our results suggest that the EZ areas became more attractive to households, as seen for example with the increases in housing costs, but the evidence also suggests that at least some of the previous literature's findings of positive effects on poverty or incomes may be due to in-migration of higher-income households to the area. This raises questions about how one should interpret the findings in the prior location-based policy literature. If the goal of the program is simply to make these economically distressed areas better or more attractive, then the program appears to have been successful. If instead, the program was supposed to aid the impoverished existing residents of the EZ areas then our results suggest that the program had little impact, despite gains in average incomes. Either way, the results of this paper suggest that interesting changes happen throughout the entire distribution and that researchers should consider measuring program effects beyond the mean.

Appendix A

See Table A1.

Table A1
Robustness checks for the effect of EZ designation on the density of population across the distribution of household income relative to the poverty level by matching method, imputed lagged variables, and bordering tracts.

Ratio of household income to poverty level	Re-weighting (base model) (1)	Re-weighting (alternate model) (2)	Nearest neighbor (3)	Imputed 1980 and 1990 block group variables (4)	EZ and EC border tracts (5)
0–0.49	0.011**	0.013*	0.012	0.009	0.057***
	[0.001, 0.024]	[−0.002, 0.029]	[−0.015, 0.029]	[−0.026, 0.052]	[0.038, 0.087]
0.50–0.74	−0.006**	−0.005	−0.005	−0.006	0.004
	[−0.012, −0.001]	[−0.011, 0.002]	[−0.014, 0.010]	[−0.018, 0.003]	[−0.002, 0.010]
0.75–0.99	−0.008***	−0.004	−0.001	−0.002	0.001
	[−0.013, −0.003]	[−0.010, 0.003]	[−0.010, 0.013]	[−0.013, 0.007]	[−0.004, 0.005]
1.00–1.24	−0.008***	−0.006**	−0.009**	−0.014**	−0.003
	[−0.012, −0.003]	[−0.011, −0.000]	[−0.020, −0.001]	[−0.030, −0.003]	[−0.006, 0.001]
1.25–1.49	−0.004*	−0.005*	−0.008	−0.001	−0.005**
	[−0.008, 0.001]	[−0.010, 0.000]	[−0.017, 0.002]	[−0.007, 0.007]	[−0.010, −0.001]
1.50–1.74	−0.002	−0.002	−0.001	0.001	−0.004*
	[−0.007, 0.002]	[−0.007, 0.003]	[−0.009, 0.009]	[−0.007, 0.007]	[−0.010, 0.000]
1.75–1.84	−0.001	−0.002	−0.003	−0.000	−0.001
	[−0.003, 0.001]	[−0.004, 0.001]	[−0.006, 0.003]	[−0.004, 0.005]	[−0.003, 0.001]
1.85–1.99	−0.001	−0.001	0.002	−0.001	−0.002*
	[−0.004, 0.002]	[−0.004, 0.003]	[−0.006, 0.007]	[−0.006, 0.004]	[−0.006, 0.000]
2.00+	0.019***	0.010	0.013	0.014	−0.047***
	[0.007, 0.030]	[−0.002, 0.024]	[−0.008, 0.034]	[−0.017, 0.040]	[−0.071, −0.030]

Notes: (1) Column (1) presents the estimates of the base model which uses propensity score reweighting separately by quintiles of 1990 block group population.
(2) Column (2) presents the estimates of an alternate propensity score reweighting procedure including controls for 1990 block group population quintiles when estimating the propensity score.
(3) Column (3) presents the estimates using nearest neighbor matching with up to 10 matches within a caliper of 0.01.
(4) Column (2) presents the using the same procedure for as the baseline model except that matching is done on imputed values for the 1980 and 1990 distributions. The imputation is done by replacing the observed distribution of households in each block with a weighted average of the distribution of block within 1 mile (using Gaussian weights within the 1 mile radius).
(5) Column (3) presents the same procedure as the baseline model except using the block groups within the tracts that directly border the EZ and EC areas.

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